

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2023
CSE-1102: Discrete Mathematics

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

- 1 a) Describe Universal and Existential quantifier with example. 4
- b) Consider Universal set U and two sets A and B . Shade the following sets using Venn diagram: 2+2
 - i. $(A \cap B)^C$
 - ii. B/A^C
- c) Among 50 students in a class, 26 got an A in the first examination and 21 got an A in the second examination. If 17 students did not get an A in either examination, how many students got A in both the examination? 2
- d) Consider the following predicates: 4

$P(x, y): x \geq y; Q(x, y): x \leq y; R(x): x - 7 = 2; S(x): x > 9$

If the universe of discourse is the natural numbers, give the truth values of each of the following propositions:

 - i. $\exists x R(x)$
 - ii. $\forall x \forall y [P(x, y) \vee Q(x, y)]$
 - iii. $\exists y \forall x [S(y) \wedge Q(x, y)]$
- 2 a) Define Propositional Logic. Which of the following sentences are Propositions? 2+2
 - i. Birds can fly.
 - ii. Answer the questions.
 - iii. $x+2=11$.
 - iv. $5+7=10$.
- b) Provide a comparison between Contraposition and Contradiction. Show that: $(p \wedge q) \wedge \neg (p \vee q)$ is a contradiction by using truth table. 2+2
- c) Find the Converse, Contrapositive, and Inverse for the following two implications: 3+3
 - i. "If it is Friday, then it is a holiday".
 - ii. "If it is raining, then there are dark clouds in the sky".
- 3 a) Define Transpose and Square matrices with example. 3
- b) Let $A=\{1,2,3,4\}$, $B=\{4,5,6\}$ and $C=\{6,7,8\}$. M is a relation from A to B and N is a relation from B to C , which is given by 4

$M = \{(x, y): x + y = 7\}$
 $N = \{(x, y): y - x = 1\}$

Determine M and N .
- c) What are the Floor and Ceiling Functions? Describe with examples. 3
- d) What is Pigeonhole Principle? Find the minimum of students in a class to be sure that three of them are born in the same month. 1+3
- 4 a) Translate the following statements into logical expressions: 4
 - i. All lions are fierce.
 - ii. Some lions don't drink coffee.

where $P(x)$, $Q(x)$ and $R(x)$ denote " x is a lion", " x is fierce" and " x drinks coffee" respectively.
- b) By using the Rules of Inference show that the premises "It is sunny this afternoon and it is colder than yesterday", "We will go swimming only if it is sunny", "If we do not go swimming, then we will take a canoe trip, and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset." 5
- c) Differentiate between Corollary and Conjecture. By using proof by Contraposition, prove that: "If n is an integer and $3n+2$ is odd, then n is odd". 5

- 5 a) Define Congruence with application. Determine whether 17 is congruent to 5 modulo 6. 2+2
 b) What is Recursion? Find the solution to the recurrence relation $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with the initial conditions: $a_0=2$, $a_1=5$ and $a_2=15$. 2+3
 c) Define One-to-One and Onto function with example. f , g , and h are functions on $X = \{1, 2, 3\}$ as:

$$\begin{aligned} f &= \{(1, 2), (2, 3), (3, 1)\} \\ g &= \{(1, 2), (2, 1), (3, 3)\} \\ h &= \{(1, 1), (2, 2), (3, 1)\} \end{aligned}$$

Compute:

1. $f \circ g \circ h$
2. $f \circ h \circ g$

- 6 a) Find the recurrence solution for the following recurrence relation using the substitution method. 4
 $a_n = a_{n-1} + 3$; $n \geq 2$ and $a_0 = 2$ for $n = 1$
 b) State and prove the principles of Inclusion and Exclusion. 4
 c) Given $A = \{1, 2, 3, 4\}$. Consider the following relation R on A . 6

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

- i. Draw the directed graph of R .
- ii. Is R Reflexive, Symmetric, Transitive or Anti-symmetric?
- iii. Find R^2 .

- 7 a) State the "Handshaking Theorem" for directed and undirected graph. How many edges are there in a graph with 10 vertices each of degree 6? 2+2
 b) Write down the necessary conditions for two graphs to be Isomorphic. Construct the Precedence Graph for the given value. 4

$$S1 a: = 0$$

$$S2 b: = 1$$

$$S3 c: = a + 1$$

$$S4 d: = b + a$$

$$S5 e: = d + 1$$

$$S6 e: = c + d$$

- c) What do you mean by Euler's Formula? Prove the theorem $V - E + R = 2$ where V , E , R means respective meaning. 1+3
 d) In how many ways can an organization containing 26 members elect a president, treasurer and secretary? 2